

Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____ 7. _____
8. _____ 9. _____ 10. _____ 11. _____ 12. _____

Warm-Up: Vertex and Axis of Symmetry

Example (Warm-Up: Vertex and Axis of Symmetry):

For $y = (x - 4)^2 + 2$, vertex is $(4, 2)$ and axis of symmetry is $x = 4$.

Directions: State the vertex and axis of symmetry for each function.

1. $y = (x - 1)^2 + 3$

3. $y = 2(x - 3)^2 + 1$

2. $y = (x + 2)^2 - 5$

4. $y = -(x + 4)^2 + 6$

Find the y-intercept

Example (Find the y-intercept):

For $y = (x - 2)^2 + 3$, substitute $x = 0$: $y = (0 - 2)^2 + 3 = 7$, so the y-intercept is $(0, 7)$.

Directions: Find the y-intercept for each function by substituting $x = 0$.

5. $y = (x - 3)^2 - 1$

7. $y = 2(x - 1)^2 - 2$

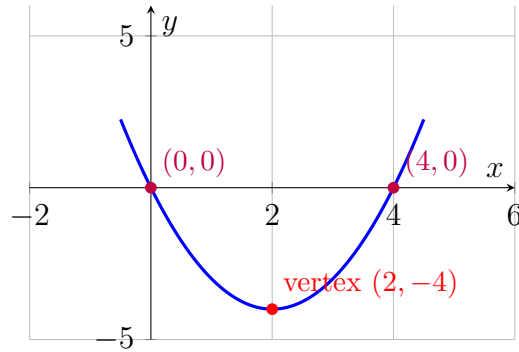
6. $y = (x + 1)^2 + 4$

8. $y = -(x + 3)^2 + 5$

Full Graphs

Example (Full Graphs):

For $y = (x - 1)^2 - 3$: vertex $(1, -3)$, axis $x = 1$, y-intercept $(0, -2)$, mirror point $(2, -2)$. Plot and connect smoothly.



Reference graph of $y = (x - 2)^2 - 4$ showing the mirror-point technique.

Directions: For each function, find the vertex, axis of symmetry, y-intercept, and its mirror point, then sketch the graph on the grid.

9. $y = (x - 2)^2 - 4$ (use the reference graph above to check your own sketch) 10. $y = (x + 2)^2 - 3$

11. $y = 2(x - 1)^2 + 1$

Challenge (same sheet):

12. Graph $y = -2(x - 1)^2 + 3$ and label all key features (vertex, axis of symmetry, y-intercept, mirror point).

Answer Key — fully worked

Procedure: Identify the vertex (h, k) from the equation.; Draw the axis of symmetry as a dashed vertical line at $x = h$.; Find the y-intercept by substituting $x = 0$ and solving for y .; Reflect the y-intercept across the axis of symmetry to get a mirror point.; Plot the vertex, y-intercept, and mirror point.; Connect the points with a smooth, U-shaped curve..

1. Vertex $(1, 3)$; axis of symmetry $x = 1$.
2. Vertex $(-2, -5)$; axis of symmetry $x = -2$.
3. Vertex $(3, 1)$; axis of symmetry $x = 3$.
4. Vertex $(-4, 6)$; axis of symmetry $x = -4$.
5. $y = (0 - 3)^2 - 1 = 8$, so $(0, 8)$.
6. $y = (0 + 1)^2 + 4 = 5$, so $(0, 5)$.
7. $y = 2(0 - 1)^2 - 2 = 0$, so $(0, 0)$.
8. $y = -(0 + 3)^2 + 5 = -4$, so $(0, -4)$.
9. Vertex $(2, -4)$, axis $x = 2$, y-intercept $(0, 0)$, mirror point $(4, 0)$ – matches the reference graph.
10. Vertex $(-2, -3)$, axis $x = -2$, y-intercept: $y = (0 + 2)^2 - 3 = 1$, point $(0, 1)$, mirror point $(-4, 1)$.
11. Vertex $(1, 1)$, axis $x = 1$, y-intercept: $y = 2(0 - 1)^2 + 1 = 3$, point $(0, 3)$, mirror point $(2, 3)$.
12. $y = -2(x - 1)^2 + 3$: vertex $(1, 3)$, axis of symmetry $x = 1$, opens down (since $a = -2 < 0$). y-intercept: $y = -2(0 - 1)^2 + 3 = -2 + 3 = 1$, point $(0, 1)$; mirror point across $x = 1$ is $(2, 1)$. Plot $(1, 3)$, $(0, 1)$, $(2, 1)$ and connect with a downward smooth curve.